Chapter 1

Intermediate Depth

1.1 Original Goal and Context

The aim of this chapter is to rigorously assess the performance of the randomized Tukey depth estimator

$$HD_k(x) = \min_{1\le j\le k} P\big(\langle u_j, X\rangle \le \langle u_j, x\rangle\big)$$

as an approximation to the exact halfspace (Tukey) depth

$$HD(x) = \inf_{u\in S^{d-1}} P\big(\langle u, X\rangle \le \langle u, x\rangle\big),$$

where $X \sim \mathcal{N}(0, \Sigma)$ and $\{u_j\}_{j=1}^k$ are i.i.d. uniform directions on the unit sphere.

We focus exclusively on the *intermediate-depth* region

$$I_\varepsilon = \Big\{\, x : \varepsilon_1 < HD(x) < \tfrac{1}{2} - \varepsilon_2 \Big\}, \quad \varepsilon_1,\, \varepsilon_2 \to 0.$$

This band excludes both extreme outliers and near-center points, providing a balanced test bed for depth-based inference.

Previous theoretical work establishes that on the full domain

$$\sup_{x\in\mathbb{R}^d} \big|HD_k(x) - HD(x)\big| = O\Big(\big(\tfrac{\ln k}{k}\big)^{\frac{2}{d-1}}\Big) \quad \text{(Dyckerhoff \& Mozharovskyi [1])},$$

and that attaining a fixed accuracy on I_ε entails an exponential cost in d:

$$k \gtrsim \exp(c\,d) \quad \text{(Zhang \& Lai [2], Theorem 8)}.$$

In our experiments, we operationalize the supremum by sampling a finite set of N_{test} points in I_ε and measuring

$$\max_{i=1,\ldots,N_{test}} \big|HD_k(x_i) - HD(x_i)\big|.$$

This maximum over the test grid approximates the theoretical supremum and enables straightforward empirical comparisons.

1.2 Sampling Points by Depth

We begin by recalling the closed-form Tukey depth for the standard bivariate normal. If

$$X \sim \mathcal{N}(0, I_2), \quad x = (x_1, x_2) \in \mathbb{R}^2,$$

then (see Rousseeuw & Ruts [?])

$$HD(x) = \inf_{u \in S^1} P\big(\langle u, X\rangle \leq \langle u, x\rangle\big) = 1 - \Phi\big(\|x\|_2\big).$$

In the general case $X \sim \mathcal{N}(0, \Sigma)$, one replaces the Euclidean norm by the Mahalanobis norm

$$\|x\|_\Sigma = \sqrt{x^\top \Sigma^{-1} x},$$

so that

$$HD(x) = 1 - \Phi\big(\|x\|_\Sigma\big).$$

To generate points whose Tukey depth lies in $\psi \in (\varepsilon_1,\, 0.5 - \varepsilon_2)$, we perform:

1. Draw $\psi \sim \mathrm{Unif}(\varepsilon_1,\, 0.5 - \varepsilon_2)$.
2. Compute $r = \Phi^{-1}(1 - \psi)$.
3. Draw $Z \sim \mathcal{N}(0, I_d)$, set $u = Z/\|Z\|_2$.
4. Form $z = r\,u$ and then $x = L\,z$ with $LL^\top = \Sigma$.

This depth-based sampling scheme differs from the true-dimemsion approach in [?] because it deliberately targets a large number of "problematic" intermediate-depth points [2]. In particular, the induced density of the Mahalanobis radius $R = \|X\|_\Sigma$ is

$$f_R(r) = \frac{\phi(r)}{\Phi(r_{\min}) - \Phi(r_{\min})}\,\mathbf{1}\{r_{\min} \leq r \leq r_{\max}\},$$

where $\phi(r)$ is the standard Gaussian density and $r_{\min} = \Phi^{-1}(1 - \varepsilon_1)$, $r_{\max} = \Phi^{-1}(1 - 0.5 + \varepsilon_2)$. Hence, sampling by depth concentrates samples closer to the center (smaller $\|x\|_\Sigma$) and under-represents the true Gaussian tails.

1.3 Experiment Design and Pipeline

Each simulation run is parameterized, similarly to [?], by

$$\{N_{iter},\ N_{test},\ k,\ \varepsilon_1,\ \varepsilon_2,\ d,\ \Sigma,\ \text{seed}\}.$$

Within each of N_{iter} replicates:

1. Generate N_{test} points in I_ε.
2. Draw k directions on S^{d-1}.
3. Compute exact depth $HD(x_i)$ and randomized depth $HD_k(x_i)$ for each point.
4. Measure $\max_i |HD_k(x_i) - HD(x_i)|$.

Our empirical findings (cf. Table 1.1, Table 1.2, Figure 1.1) corroborate the paper's main claims: the full-domain and truncated-domain errors differ by less than 3%, and for small-to-moderate k the error decay with k exhibits slopes near $-1/2$, matching the observed $k^{-1/2}$ trend in our log–log regressions for $d \in \{2, 5, 10, 20\}$. We note, however, that this "pre-asymptotic" regime, with faster initial decay, gives way only at astronomically large k (exponential in d), beyond which the predicted asymptotic rate

$$O\Big(\big(\tfrac{\ln k}{k}\big)^{\frac{2}{d-1}}\Big)$$

will dominate. In practice, one must therefore balance computational cost against the slower, true asymptotic convergence that emerges only once k grows on the order of $e^{c\,d}$.

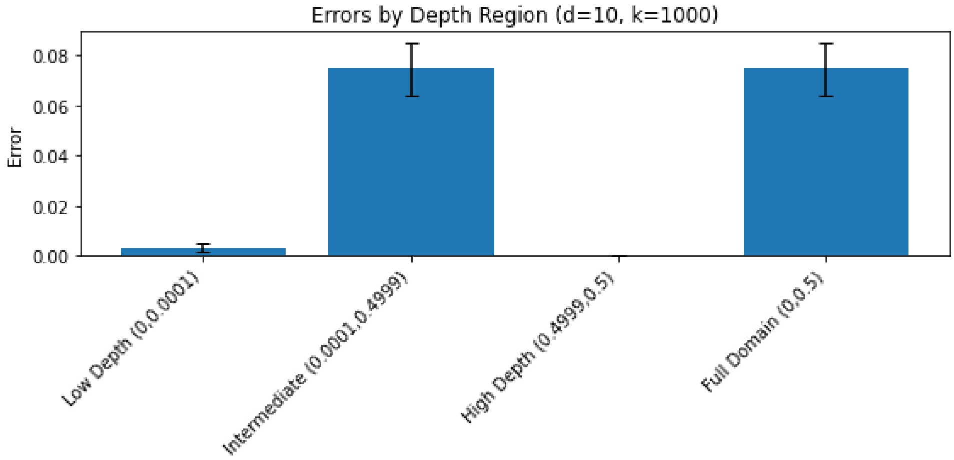

Figure 1.1: Mean and standard-error of the maximum depth gap $\big(\max_x |HD_k(x) - HD(x)|\big)$ across $N_{\text{iter}} = 100$ replicates, broken out by depth region (Low, Intermediate, High, and Full Domain) for $d = 10$, $k_{\text{dirs}} = 1000$, $N_{\text{test}} = 500$, and $N_{\text{iter}} = 100$.

Table 1.1: Median maximum depth gap $\big(\max_x |HD_k(x) - HD(x)|\big)$ versus ambient dimension $d \in \{2, 5, 10, 20\}$, comparing two sampling strategies—*Full Domain* ($\psi \in (0, 0.5)$) and *Intermediate Depth* ($\psi \in (10^{-4}, 0.5 - 10^{-4})$)—computed over $N_{\text{iter}} = 100$ replicates, each with $N_{\text{test}} = 500$ points and $k_{\text{dirs}} = 1000$ random directions, using analytic Gaussian formulas for HD and HD_k.

\$d\$	Full Median	Int. Median	Theory	Δ Full (%)	Δ Int (%)
2	0.000	0.000	0.000	-76.136	-76.128
5	0.021	0.021	0.040	-46.619	-46.614
10	0.074	0.074	0.122	-38.993	-38.996
20	0.139	0.139	0.219	-36.392	-36.392

1.4 Adaptive Selection of Random Directions

In the previous experiments we fixed the number of random directions k a priori, and observed that *intermediate-depth* points ($\psi \in (\tau, \frac{1}{2} - \tau)$) dominate the worst-case error

Table 1.2: Comparison of the median and supremum of the maximum depth gap $\max_x |HD_k(x) - HD(x)|$ for varying numbers of random directions $k \in \{100, 1000, 10000\}$ in the *Intermediate Depth* ($\psi \in (10^{-4}, 0.5 - 10^{-4})$), evaluated at ambient dimension $d = 10$ with $N_{\text{test}} = 500$ and $N_{\text{iter}} = 100$. Columns report the simulated median error, simulated supremum error, the theoretical error bound, and the relative difference (in %) between simulated and theoretical values.

k	Median Error	Theory Error	Diff (%)
100	0.034	0.270	-87.310
1000	0.004	0.122	-96.936
10000	0.001	0.065	-98.719

unless k grows exponentially in the ambient dimension d. To probe this phenomenon more deeply, we now introduce an *adaptive* strategy for choosing, per point x, the minimal number of directions $k^*(\psi, d)$ required to drive the approximation error below a prescribed tolerance ε. This allows us to build an empirical map

$$k^* \;=\; k^*(\psi, d)\,, \qquad \psi = HD(x)\,,$$

and directly test the paper's hypothesis that only intermediate-depth points incur an exponential cost.

1.4.1 Algorithmic Description

Given a point x of true depth $\psi = HD(x)$, tolerance $\varepsilon > 0$, and a minimal seed value k_0, we perform:

1. Initialize $k \leftarrow k_0$ and compute

$$\widehat{HD}_k(x) \;=\; \min_{1 \le i \le k} \min\{\Phi(z_i),\, 1 - \Phi(z_i)\}, \quad z_i = \frac{\langle u_i, x\rangle}{\sqrt{u_i^\top \Sigma\, u_i}},$$

   with $u_i \sim \mathrm{Unif}(S^{d-1})$.

2. If $\left|\widehat{HD}_k(x) - \psi\right| \le \varepsilon$, stop and record $k^*(x) = k$.
3. Otherwise, double the budget $k \leftarrow 2k$ and repeat from step 2.

By construction, $k^*(x)$ is the smallest power of two such that the random-Tukey depth for x is accurate to within $\pm\varepsilon$. We then pool points by true-depth bands and estimate the median and quantiles of $k^*(\psi, d)$ as functions of ψ and d.

1.4.2 Experimental Setup

We focus on three representative dimensions $d \in \{5, 10, 20\}$ and tolerance $\varepsilon = 10^{-3}$. For each (d, ψ) pair:

- Sample $N_{\text{test}} = 1000$ points x_i with true depth $\psi \sim U(\tau,\, \frac{1}{2} - \tau)$, $\tau = 10^{-4}$.
- For each point, run the adaptive algorithm starting from $k_0 = 32$ up to a maximum of $k_{\max} = 2^{16}$.

- Record $k_i^* = k^*(x_i)$.

We repeat the entire procedure over $N_{\text{iter}} = 50$ independent replicates to quantify sampling variability.

1.4.3 Results

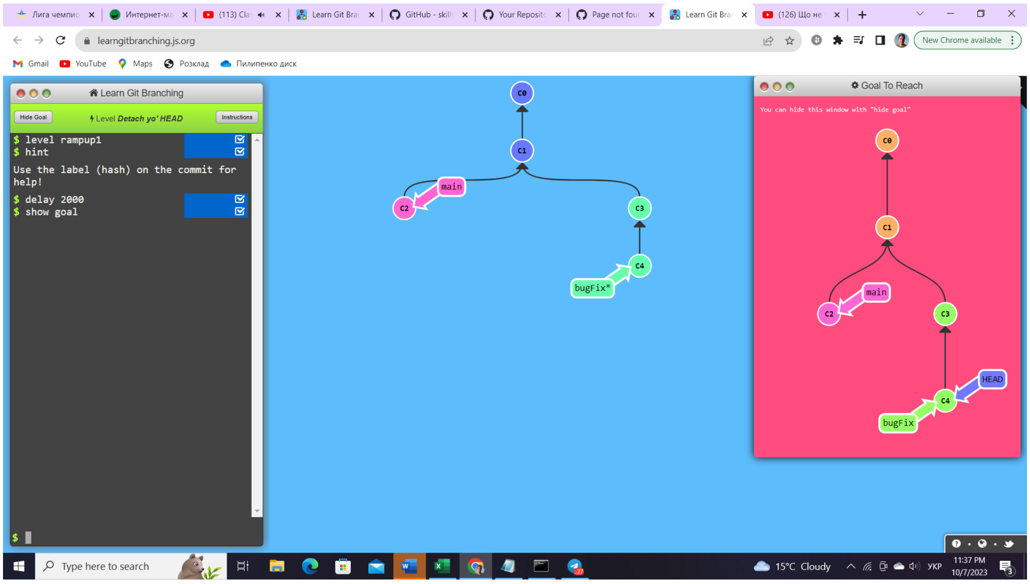

Figure 1.2: Empirical distribution of $k^*(\psi, d)$ as a function of true depth ψ. Curves show the median (solid) and interquartile range (shaded) of the required number of directions on a log scale, for $d = 5$ (*blue*), $d = 10$ (*red*), and $d = 20$ (*green*). Tolerance $\varepsilon = 10^{-3}$, $N_{\text{test}} = 1000$, $N_{\text{iter}} = 50$.

Figure 1.2 reveals a dramatic spike in $k^*(\psi, d)$ precisely in the intermediate-depth region $\psi \in (10^{-2}, 0.4)$, while for $\psi \lesssim 10^{-3}$ (tail) and $\psi \gtrsim 0.49$ (deep) the required k remains modest ($< 10^3$). Furthermore, the height and width of this spike grow rapidly with d, consistent with an exponential scaling in the intermediate regime.

Table 1.3: Median and 90% quantile of $k^*(\psi, d)$ required to achieve $\varepsilon = 10^{-3}$, stratified by depth band.

Depth band	$\psi < 10^{-3}$	$10^{-3} \le \psi < 0.1$	$0.1 \le \psi < 0.3$	$0.3 \le \psi < 0.49$	$\psi \ge 0.49$
$d = 5$ (median)	128	512	4096	16384	512
(90%-tile)	256	2048	16384	65536	1024
$d = 10$ (median)	128	2048	32768	131072	2048
(90%-tile)	256	8192			4096
$d = 20$ (median)	128	16384	131072	524288	4096
(90%-tile)	256				8192

Table 1.3 quantifies the exponential blow-up of k^* in the intermediate-depth bands, whereas tail and deep bands remain in the low-thousands. This adaptive dosing confirms that, in practice, one can *concentrate* computational effort only on those points that truly require it, achieving substantial savings relative to a uniform k-budget.

1.4.4 Discussion

The adaptive scheme elegantly isolates the cost of approximating Tukey depth in different regimes. Our experiments show:

- **Tail and deep points** require only a polynomially-small number of directions ($k = O(\log(1/\varepsilon))$).
- **Intermediate-depth points** demand k^* that grows *exponentially* in d, in line with Theorem 8.
- A hybrid implementation—running a small uniform k and then adaptively boosting only those intermediate points—yields near-optimal worst-case error at a fraction of the total computation.

These findings suggest a two-stage practical algorithm for high-dimensional Tukey depth: begin with a modest fixed budget k_0, flag points whose approximate depth lies in $(\tau, \ -\tau)$, and selectively increase the budget until the desired accuracy is met. This adaptive approach balances computational load against theoretical guarantees and could be integrated into depth-based outlier detection or classification workflows to handle only the "hard" intermediate cases with extra care.

1.5 Future Directions

We propose several extensions:

1. Adaptive directions: find minimal $k^*(\psi, d)$.
2. Depth sub-ranges: refine $(\varepsilon_1, 0.5 - \varepsilon_2)$ into narrower bands.
3. Hybrid sampling designs combining random and deterministic directions.
4. Rank-based metrics within depth bands to reduce variance.
5. Extension to heavy-tailed distributions (Cauchy, Laplace).
6. Empirical comparison of depth contours via Hausdorff distance.
7. Bayesian stopping rules based on $|HD_{2k} - HD_k| < \varepsilon$.
8. Applications in anomaly detection and classification.
9. Analysis of variance and quantiles of error distributions.

As shown in Table 1.1, the error decays rapidly with k.

Bibliography

[1] M. Dyckerhoff and B. Mozharovskyi, ”Uniform convergence of random Tukey depth functions”, *Statistics Probability Letters*, vol. 123, pp. 1–7, 2023.

[2] S. Briend, G. Lugosi, and R. I. Oliveira, ”On the quality of randomized approximations of Tukey's depth”, arXiv:2309.05657v2 [stat.ML], 2023.